to reproduce classical mechanics in circumstances where it should—where the particles are heavy, the potentials are smooth, and nothing causes the wave function to disintegrate or scatter.[6]

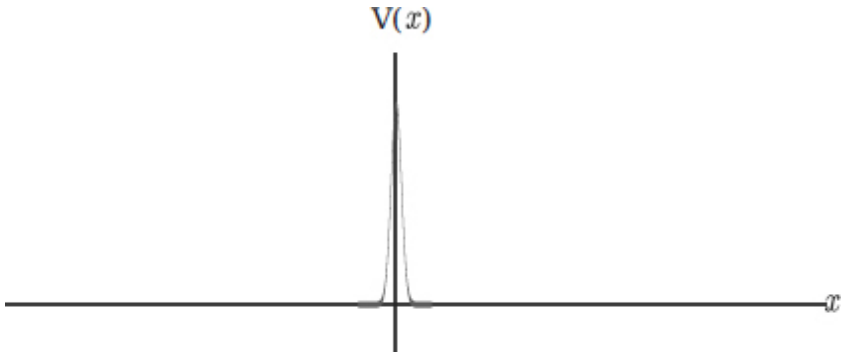


Figure 9.4: Spiky Potential Function. Potential functions with sharp peaks tend to cause wave functions to scatter. The smaller these features are in relation to the wave packet, the more the wave packet will scatter, and the less "classical" it will become.

What physical situations lead to "bad potentials" that break up the wave function? Suppose a potential has features that have a certain size associated with them. Think of Fig. 9.4 on steroids, with lots of large, closely packed spikes. Suppose we call the size of these features δx, and that δx is significantly smaller than the incoming particle's uncertainty in position:

$$\delta x < \Delta x.$$

If the sharp features of $V(x)$ exist on a scale that is much smaller than the size of the incoming wave packet, the packet will break into a lot of little pieces. Each one will scatter off in a different direction. Roughly speaking, when the features of the potential are shorter than the wavelength of the incoming particle, the wave function will tend to break up.

Let's say you take a bowling ball and ask, "What is Δx?" We can use the uncertainty principle to gain some intuition about this question. Typically, $\Delta p \times \Delta x$ is bigger than $\hbar$. But in many reasonable cases it's of order $\hbar$:

$$\Delta p \Delta x \sim \hbar.$$